

SCHAFFHAUSEN, Switzerland

WMO: 066200

Lat: 47.6897N

Lon: 8.6200E

Elev: 1434

StdP: 13.95

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.8	20.2	8.2	8.8	18.6	14.0	11.8	23.8	31.3	44.2	26.9	43.8	7.7	50	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	87.6	67.4	84.0	66.3	80.7	65.1	69.2	82.9	67.8	80.4	66.4	77.9	7.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.9	97.1	73.4	63.5	92.6	72.0	62.4	88.8	70.7	34.1	83.1	32.9	80.7	31.8	78.0	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.4	20.5	17.1	DB	13.3	92.7	5.5	2.8	9.4	94.7	6.1	96.3	3.0	97.9	-1.0	99.9	(4)
(5)			WB	11.8	72.1	5.9	2.0	7.6	73.5	4.2	74.7	0.9	75.8	-3.4	77.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	33.6	35.3	43.0	51.0	57.6	65.0	67.9	66.7	59.2	51.1	41.3	34.7	(6)	
	DBStd	13.97	7.38	7.52	7.19	7.04	6.62	6.72	5.89	5.98	5.94	6.56	6.30	6.61	(7)	
	HDD50	2056	508	413	233	73	12	1	0	0	4	68	270	474	(8)	
	HDD65	5593	972	832	683	420	243	83	39	50	189	431	712	939	(9)	
	CDD50	2285	1	1	15	104	248	451	554	519	281	103	8	0	(10)	
	CDD65	347	0	0	0	0	14	84	128	104	17	0	0	0	(11)	
	CDH74	2907	0	0	0	21	150	729	1076	821	109	1	0	0	(12)	
	CDH80	867	0	0	0	2	22	216	350	267	10	0	0	0	(13)	
(14) Wind	WSAvg	7.5	8.2	8.5	8.7	7.5	7.5	7.1	6.9	6.5	6.8	6.8	7.4	8.1	(14)	
(15) Precipitation	PrecAvg	39.4	2.8	2.6	2.6	2.9	4.1	4.1	4.3	3.7	2.9	3.0	3.1	3.3	(15)	
	PrecMax	52.0	6.6	6.7	8.9	6.5	7.8	7.0	7.6	6.9	5.8	5.5	6.7	7.1	(16)	
	PrecMin	29.9	0.4	0.6	0.6	0.3	0.9	1.5	1.3	0.9	1.1	0.2	0.1	0.1	(17)	
	PrecStd	5.7	1.7	1.5	1.8	1.5	1.8	1.4	1.6	1.5	1.2	1.5	1.6	1.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.5	57.6	67.6	77.0	82.8	90.0	92.3	92.0	81.2	70.8	61.5	52.2	(19)	
		MCWB	48.6	47.3	51.8	57.3	63.2	68.8	68.4	67.5	64.6	60.1	53.2	48.5	(20)	
	2%	DB	50.1	52.5	63.1	72.5	78.1	85.8	87.6	86.5	77.3	66.5	55.9	49.3	(21)	
		MCWB	46.3	44.8	49.4	54.9	61.4	66.8	67.0	67.7	63.6	58.4	51.4	46.6	(22)	
	5%	DB	46.8	49.2	58.8	68.7	74.2	81.9	83.8	82.4	73.0	63.0	52.7	46.6	(23)	
		MCWB	43.5	43.3	48.0	53.3	59.2	65.7	66.0	66.7	62.0	56.8	49.2	44.3	(24)	
	10%	DB	43.5	45.4	54.8	64.6	70.1	78.0	80.2	78.3	69.4	60.1	50.0	43.9	(25)	
		MCWB	41.2	41.0	46.5	51.6	57.7	64.3	64.9	65.2	60.5	55.5	47.0	41.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.0	48.5	53.3	58.7	65.5	71.1	70.8	70.9	67.1	61.8	54.4	49.4	(27)	
		MCDB	52.4	55.4	62.7	73.6	78.4	85.4	86.6	84.3	77.0	68.0	59.6	51.3	(28)	
	2%	WB	46.6	45.8	50.9	56.3	63.2	68.8	68.9	69.1	65.0	59.6	51.8	47.0	(29)	
		MCDB	49.4	51.3	59.1	69.3	75.1	82.1	82.8	82.4	74.1	64.5	55.0	49.1	(30)	
	5%	WB	43.7	43.8	49.1	54.3	61.0	66.8	67.4	67.5	63.2	57.8	49.8	44.6	(31)	
		MCDB	46.4	48.7	56.9	65.7	70.9	79.1	80.1	79.7	71.0	62.0	52.7	46.4	(32)	
	10%	WB	41.3	41.4	47.0	52.5	58.9	65.1	66.0	66.0	61.4	56.0	47.4	42.0	(33)	
		MCDB	43.4	45.0	54.2	62.8	67.6	76.0	77.4	76.5	67.9	59.6	49.6	43.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	11.2	15.8	18.2	17.5	18.2	18.6	17.8	16.2	13.1	9.1	7.5	(35)	
		MCDBR	11.8	18.0	24.3	26.0	24.6	24.8	25.2	24.7	21.7	17.7	13.7	10.2	(36)	
	5% WB	MCWBR	8.9	11.3	12.8	12.0	10.8	10.0	9.4	9.3	10.0	10.1	9.3	7.9	(37)	
		MCWBR	10.9	16.2	20.8	23.1	21.6	22.6	22.3	22.4	19.1	16.0	12.8	9.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.299	0.325	0.360	0.392	0.405	0.409	0.410	0.404	0.379	0.354	0.322	0.296	(40)	
		taud	2.447	2.370	2.326	2.264	2.270	2.298	2.288	2.321	2.370	2.425	2.467	2.453	(41)	
	Ebn at Noon		249	265	271	272	272	269	266	262	250	238	235		(42)	
		Edh at Noon	21	28	34	40	41	40	37	32	26	21	18		(43)	
(44) All-Sky Solar Radiation	RadAvg		350	610	993	1366	1586	1817	1743	1523	1137	709	390	286	(44)	
	RadStd		39	86	134	195	161	165	158	156	118	76	47	45	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	+0.83	+0.58	N/A	N/A	N/A	N/A	(46)
(47)	Regional (114 neighbors)	N/A	N/A	N/A	+1.26	+0.76	+0.61	N/A	N/A	+101	+25	(47)

Nomenclature: See separate page